

DIKSHA SINGH

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PROFESSIONAL SUMMARY

BCA student and Founder of NXIO — an AI-powered algorithmic trading intelligence platform for NSE/BSE/MCX markets. Skilled in machine learning, data analysis, EDA, feature engineering, NLP/sentiment analysis, and full-stack API development. Built 5 production-grade ML projects spanning LSTM signal generation, FinBERT news sentiment engine, label noise detection, prediction confidence monitoring, and REST API development. Seeking a Data Analyst, ML Engineer, or Analytics Internship role.

TECHNICAL SKILLS

Languages: Python (Advanced), SQL, Bash

ML / AI: Scikit-learn, TensorFlow, Keras, XGBoost, LSTM, HuggingFace Transformers, FinBERT, SHAP, Model Calibration, Time-Series Analysis, Statistical Hypothesis Testing, A/B Testing, Feature Engineering, EDA

Data & Visualization: Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly, Microsoft Power BI

Backend & Infra: FastAPI, PostgreSQL, SQLite, Streamlit, SQLAlchemy, JWT Auth, REST APIs, Docker, Git, GitHub

Domain: Algorithmic Trading, NSE/BSE/MCX, SEBI Compliance, ML Observability, Data Quality Engineering, Backtesting, Business Intelligence

WORK EXPERIENCE

Founder & Lead Data Scientist | NXIO — Algorithmic Trading Intelligence Platform

Oct 2025 – Present

- Designed and deployed hybrid LSTM + indicator signal engine (RSI, MACD, EMA, VWAP) across 4 timeframes (1M/5M/15M/1H), achieving ~65–68% signal accuracy vs ~52% raw LSTM baseline across 9 trained models; SEBI April 2026 Algo-ID compliant.
- Integrated HuggingFace FinBERT sentiment analysis to cross-validate LSTM trade signals against financial news headlines, reducing conflicting signal noise — deployed as Streamlit UI on GitHub.
- Built multi-exchange feature engineering pipelines (NSE/BSE/MCX), paper trading engine with virtual INR 1,00,000 capital, real-time P&L tracking, FII/DII flow detection, and bilingual voice alerting.

Independent ML Researcher | ML Reliability & Data Quality — Self-Directed

Jun 2024 – Present

- Prototyped 3 research tools targeting label noise detection, prediction calibration decay, and feature importance instability; delivered with formal documentation, ablation studies, and reproducible experiment designs.
- Applied statistical hypothesis testing, A/B testing, and SHAP-based model interpretability to validate findings and ensure production-readiness.

PROJECTS

Label Quality & Noise Engine (LQNE) | Python, Scikit-learn, Pandas

Dec 2025 – Feb 2026

- Detected mislabelled samples via cross-validation confidence thresholding and ensemble disagreement scoring; improved model accuracy from 0.91 to 0.95 on a 41,188-sample dataset; validated across 10/20/30% synthetic noise levels; open-sourced on GitHub.
- Designed ablation studies comparing 4 noise detection methods; documented reproducible experiment pipeline with full statistical validation and performance benchmarking reports.

Prediction Confidence Decay Tracker | FastAPI, Streamlit, PostgreSQL, SciPy

2025

- Full-stack ML calibration monitoring system with JWT auth, Brier score analytics, Plotly ECE/reliability curve dashboard, automated drift alerts, 11 REST API endpoints with complete Swagger documentation.
- Implemented time-windowed Brier score tracking to detect model confidence degradation over time; enabled automated email alerts when calibration drift exceeded defined thresholds.

Feature Importance Instability Detector | Python, Scikit-learn, SHAP, NumPy

2025

- Diagnostic tool flagging instability in SHAP and permutation-based feature rankings across bootstrap samples — enabling detection of unreliable features before production deployment.
- Benchmarked feature rank variance across 100 bootstrap iterations; reduced false-positive feature selection rate by identifying top unstable features responsible for model performance degradation.

HACKATHONS & CERTIFICATIONS

- E-Raksha Hackathon | IIT Delhi — National Cybersecurity Competition (2026)
- Bharatiya Antariksh Hackathon | ISRO, Powered by Hack2Skill — Acknowledged for Innovation (2025)
- redBus Data Decode Hackathon | Demand Forecasting, RMSE-evaluated leaderboard (2025)
- Microsoft Power BI | UniAthena x Cambridge International Qualifications — Blockchain Verified (Jan 2026)

EDUCATION

Bachelor of Computer Applications (BCA) — CCS University, Meerut

2023 – Present

Coursework: Data Structures, Algorithms, DBMS, Computer Networks, Mathematics